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Trust, Types & Transparency: TIC-TAC-TOE

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Trust **I**n **C**ontext

- Beyond binary: multiple authorities

Types **A**nd **C**ategories

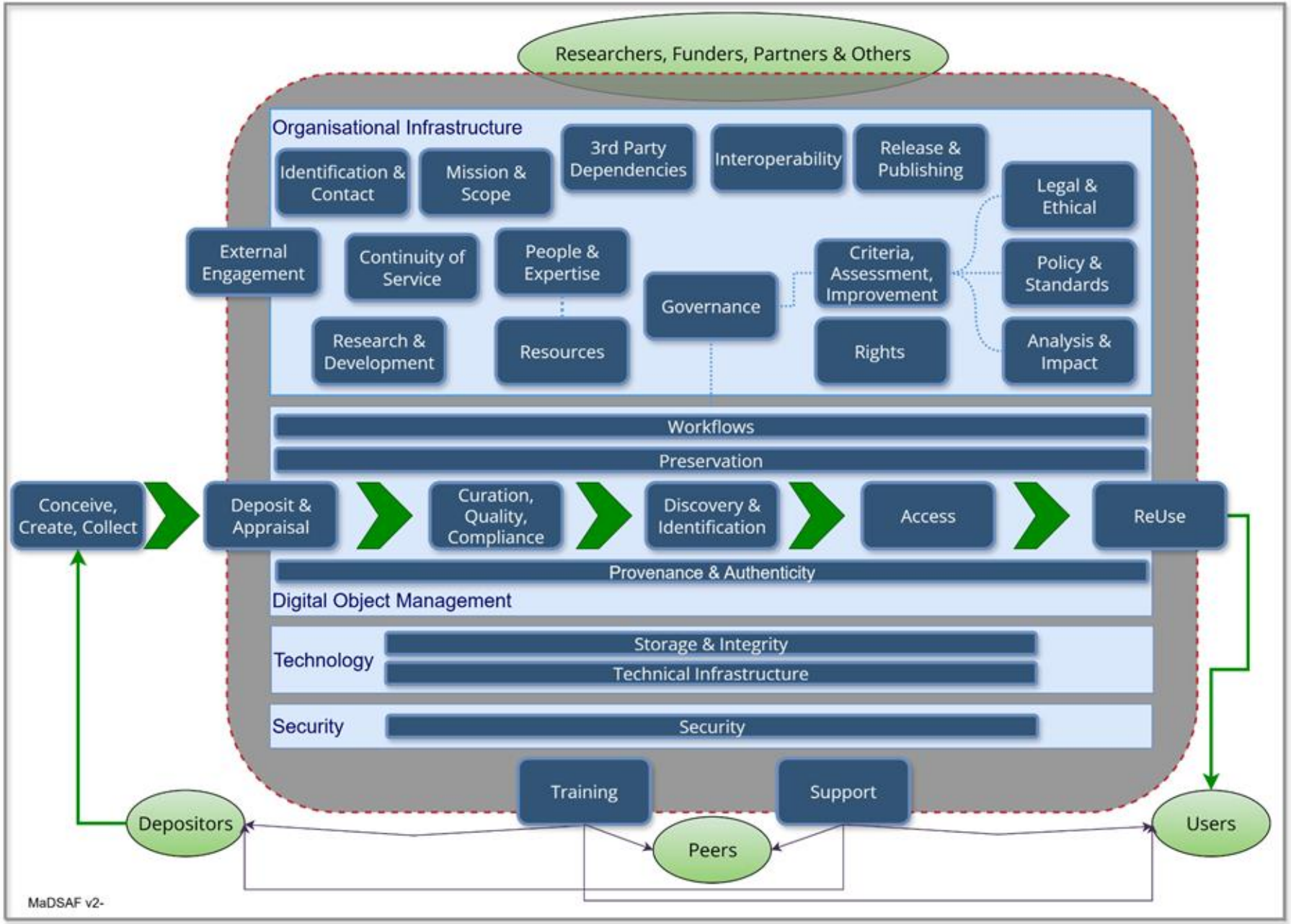
- Beyond repository types: users, depositors, objects

Transparent **O**bjects and **E**ntities

- Organisations, catalogues, data and metadata

FIDELIS Project: towards a European Network of Trustworthy Repositories

01	Set up, develop, and operate a European network of trustworthy repositories that will support the development and growth of TDRs within the EOSC ecosystem	• Set up a lean governance structure to establish and operate the Network • Build up a good membership base • Develop and test the value proposition of the Network • Provide common directionality and a common voice for digital repositories • Develop a common understanding of legal requirements of TDRs • Develop strategic partnerships and agreements with other initiatives
02	**Foster harmonisation and interoperability across repositories to enable an EOSC federation of TDRs**	• **Develop matrix of repository capabilities and characteristics** • Recommend standards, APIs and solutions to support access, interoperability and cross-repository services • Facilitate technical federation of repositories across EOSC
03	Strengthen the upskilling of repositories and expansion of the Network through an active training and support programme	• Identify training needs and support requests from digital repositories • Develop a comprehensive training and mentoring programme • Onboard new repositories in the Network • Disseminate and demonstrate the added value of the Network and its activities across EOSC



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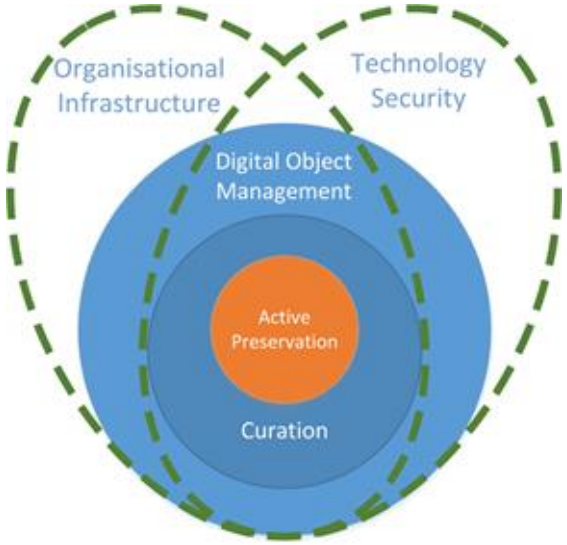
Transparent, Trustworthy Repository Attributes Matrix: TTRAM



'Superset' of 30 activities and functions

- digital object management
- organisational infrastructure
- technology
- security

Drawn from ***existing*** criteria, guidelines, requirements and standards – *not* intended to dilute or replace them.

Should complement and support those developing more specific and granular criteria.

Transparent, Trustworthy Repository Attributes Matrix: TTRAM

Purpose:

- **Collecting** information in a logical, reusable way
- **Understanding** the wide variety of types and practices across repositories
- **Sharing** structured information for consultation and use
- **Designing** and **implementing** a functional network of mutual support and aligned practice

However… **context matters**

Trust is not a binary and static outcome of a single assessment.

Defining trustworthiness **depends on the context** of objects, actors, roles, and expected levels of service.

Assessment of trustworthiness needs to be **based on reliable, transparent information**.

Examples of *different* assumptions and expectations

- CoreTrustSeal
  → active preservation repositories
- European Commission
  → funded project outputs
- European Open Science Cloud (EOSC)
  → engaged with EOSC Nodes

Trust Authorities

Identify and understand key actors in terms of:

- Criteria
  - authorship, change, versioning and release.
- Assessment Process
  - evaluation methods, outcomes, terms and how they change.
- Status
  - how it is assigned and how it might be revoked.

TAC: Types And Categories

Repository **types** commonly referred to include:

- Institutional
- Regional, National, International
- Generalist
- Disciplinary/Domain/Specialist

Influence **scope** of repository including:

- Digital objects types and content accepted
- Range of users served
- Range of depositors served
- Skills and expertise needed

TAC: Types And Categories

"Where can I (researcher Type X) ***deposit*** *(digital object Type A)"*

"Where can I (researcher Type Y) ***access*** *(digital objects Type B)"*

Forte: from Generic to Specialist

Levels of Curation and Preservation (LoRCAP)

- Variables of activities and functions (A|F in TTRAM).
- Often relevant to Trust
- Demonstrated and assessed via transparent information

TOE: Transparent Objects and Entities

Trust Targets

Understand scope of entities we are examining

We refer to 'repositories' in the broadest sense, but any venue where a user (human or machine) interacts with digital objects' data and metadata is in scope.

Digital Object Venue Entity

(DOVE)

TOE: Transparent Objects and Entities

Digital Object Venue Entity
(DOVE)

- **Organisation**(s) responsible for
- **Catalogue**(s) listing
- **Digital Objects**

Transparent information about these entities is implied by their: Coverage, forte, levels of curation, and A|F

TIC-TAC-TOE

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Transparent **O**bjects and **E**ntities

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Now and Next?

- Trustworthy Digital Repository (TDR) status is assigned based on different criteria – trust is contextual rather than binary.

- Type and level of retention, curation and preservation are insufficient to meaningfully categorise repositories

- Understanding of context *and* 'types' enables better recommendations about which information should be transparent to assess trustworthiness.

Now and Next?

Kleemola, M., L'Hours, H., Parkes, O., Recker, J., Duvaud, S., van Horik, R., Alaterä, T. J., Liberante, F., Conzett, P., Kaartinen, H., Bäckman, S., Lammert, A., & Verburg, M. (2025). FIDELIS MS12 **First version of TTRAM**. Zenodo. https://doi.org/10.5281/zenodo.17157958

Further work on

- Digital Object Venue Entities (DOVEs)
- Authority models
- Depositor, user and object type characteristics

L'Hours, H., Parkes, O., Kleemola, M., & Liberante, F. (2026). **Trust, Types, Transparency. TIC-TAC-TOE** (v01.00). Zenodo. https://doi.org/10.5281/zenodo.18640493

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Thank you